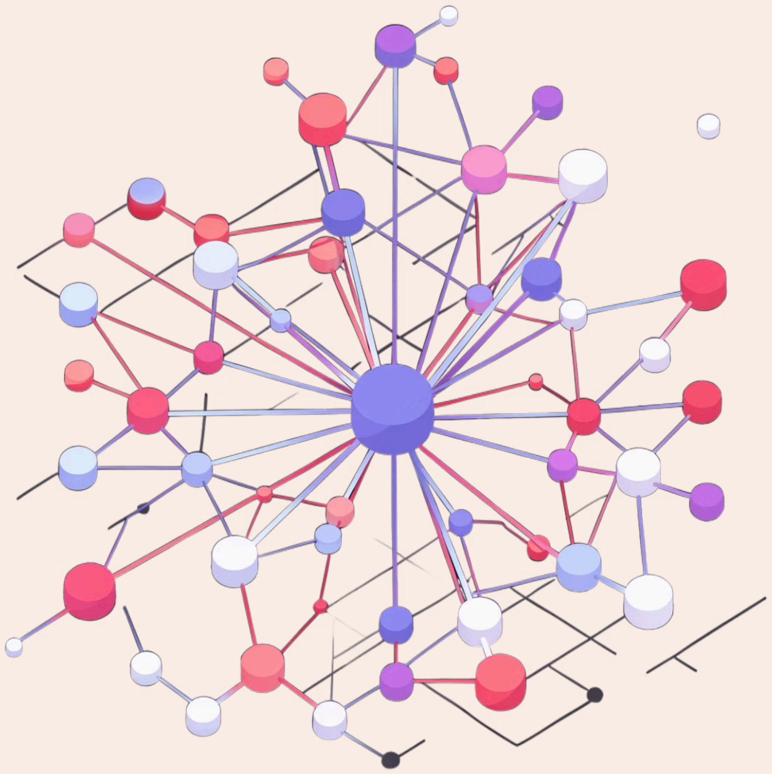
Building a Diffusion Model from Scratch

Denoising Diffusion Probabilistic Models (DDPM) on CIFAR-10

Reference: Ho et al. "Denoising Diffusion Probabilistic Models."
NeurIPS 2020. arXiv:2006.11239



The Problem: Why is Image Generation Hard?

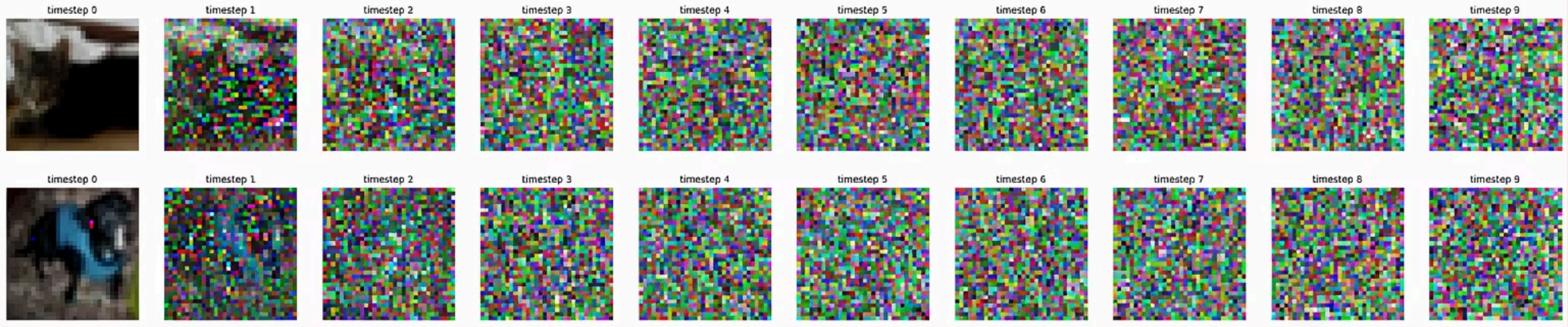
Standard generative models struggle to capture the full complexity of real image distributions. Diffusion models solve this by learning to reverse a gradual noising process — turning pure Gaussian noise back into a realistic image through hundreds of small denoising steps.

🎯 Project Goal

Implement DDPM entirely from scratch in PyTorch, train on a single CIFAR-10 class, and demonstrate image generation from random noise.

🚫 No Shortcuts

No pre-trained weights. No HuggingFace pipelines. Built from the ground up.



The Forward Process: Destroying Information Slowly

The forward process gradually corrupts a clean image x_0 by adding Gaussian noise over **T=300 timesteps**, controlled by a linear beta schedule (β from 0.0001 to 0.02).

Key Insight — The Reparameterization Trick

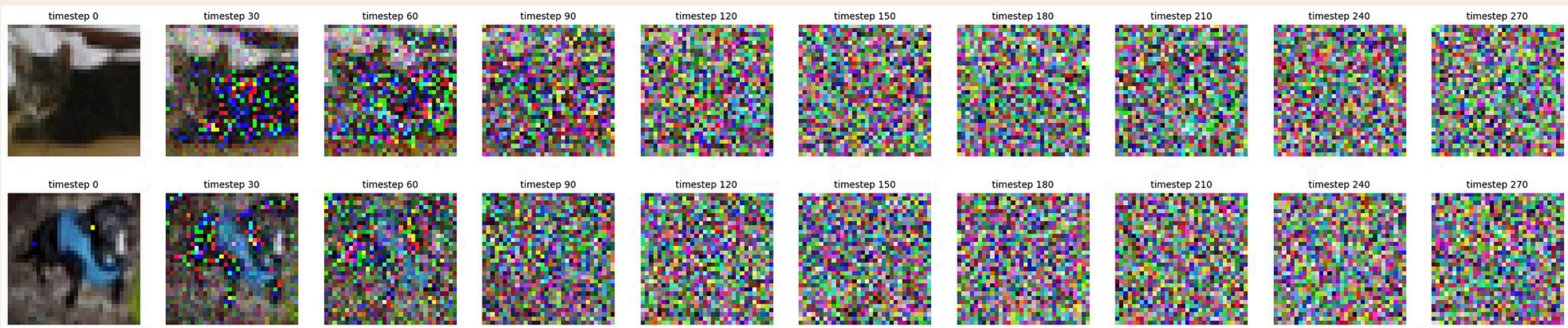
Lets us jump to **ANY timestep in one step**:

$$q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot x_0,\ (1-\bar{\alpha}_t) \cdot I)$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \varepsilon$$

No need to iterate 300 times — one matrix multiply gets us to any noise level directly.



The Reverse Process + U-Net

The reverse process learns to undo the noise step by step. A U-Net ε_θ is trained to predict the noise added at each timestep.

Reverse Sampling Step (Ho et al. Eq. 11)

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon_\theta(x_t, t)\right) + \sqrt{\beta_t}\,z$$

where $z \sim \mathcal{N}(0, I)$ for $t > 0$, and $z = 0$ at $t = 0$ (no noise at final step).

Training Objective (Ho et al. Eq. 14)

Minimize MSE between predicted and actual noise:

$$L = \mathbb{E}\left[\|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon,\ t)\|^2\right]$$

U-Net Architecture

5-Level Encoder-Decoder

Skip connections throughout

Channels

64 → 128 → 256 → 512 → 1024

Timestep Embedding

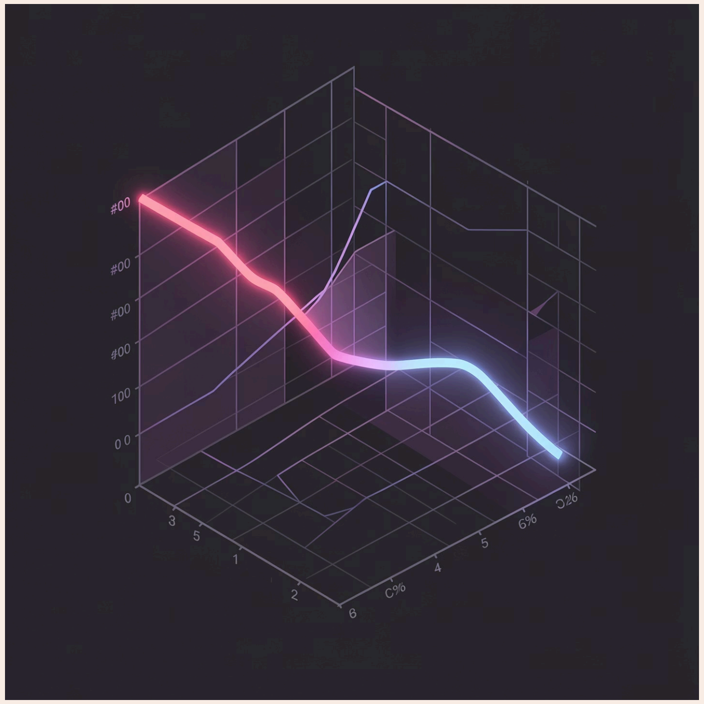
Sinusoidal embeddings inject t at every block

62.7M Parameters

Total model capacity

Made with GAMMA

Training Setup and Results



Dataset & Training Config

- **Dataset:** CIFAR-10 automobile class – 5,000 training images, 32×32 RGB, normalized to [−1, 1]
- **Training:** 500 epochs, Adam optimizer (lr=1e-3), Google Colab A100 GPU
- **Key Improvement – EMA:** Exponential Moving Average (decay=0.995) applied to model weights – significantly reduces checkerboard artifacts and produces smoother generated images

Loss dropped from **~1.0** at step 1 to **~0.05** at convergence – healthy training with no instability.

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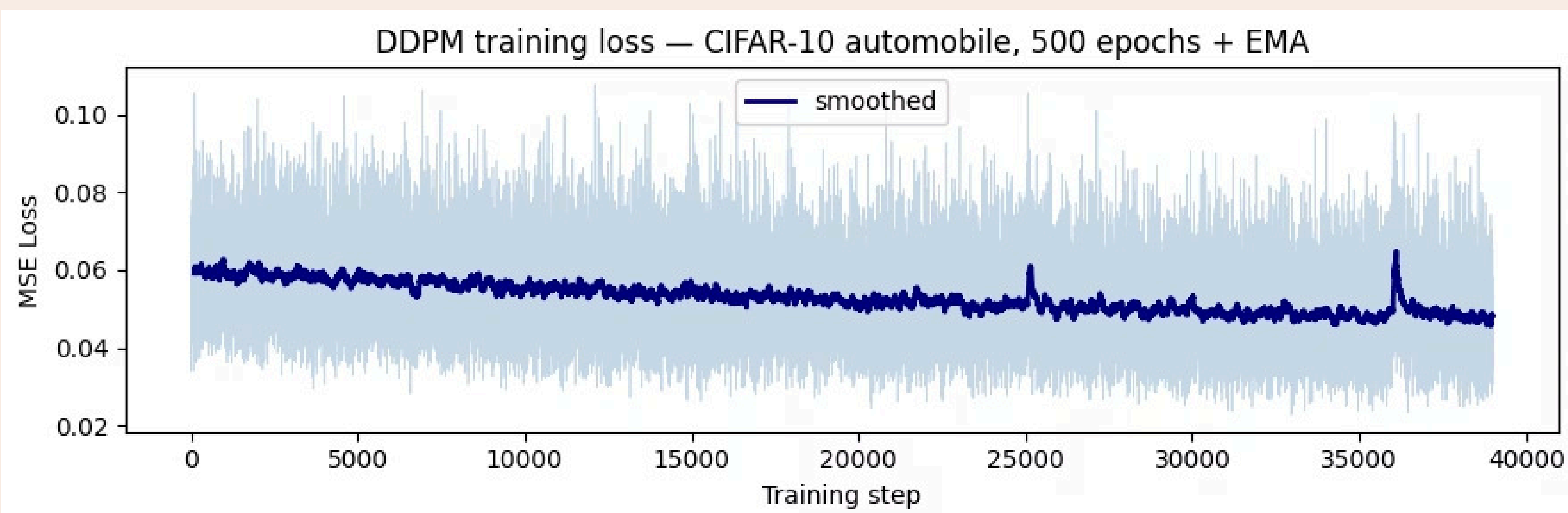

Generated Images

Starting from $x_T \sim \mathcal{N}(0, I)$, the trained U-Net denoises over **T=300 steps** to produce new automobile images.

Recognizable Structure

All 8 generated images show recognizable automobile structure – correct horizontal orientation, realistic color palettes (blues, silvers, reds), visible body and window shapes, background road context.

Resolution Limitation

At 32×32 resolution some blurriness is inherent to DDPM at this scale – this is **not** a training failure but a resolution limitation.

Quality Improves at Scale

Quality improves significantly with larger architectures and higher resolution (as seen in Stable Diffusion, DALL-E).

DDPM generated automobiles — 500 epochs, EMA

Denoising Trajectory

The reverse trajectory from $t = 299 \to t = 0$ shows how the model incrementally recovers structure:

1. **t = 299 → 200**

   Pure high-frequency noise — model making global color decisions

2. **t = 150 → 100**

   Low-frequency structure emerges — color regions start clustering

3. **t = 50**

   Clear car-like shape visible — horizontal body, darker window regions

4. **t = 20 → 0**

   Fine details sharpen — background separates from foreground

This matches exactly the expected behavior from Ho et al. — **coarse global structure forms first, fine details are added last.**

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Connection to Part I + Key Takeaways

This DDPM (Part II) is exactly the **slow teacher model** that one-step methods from Part I compress.

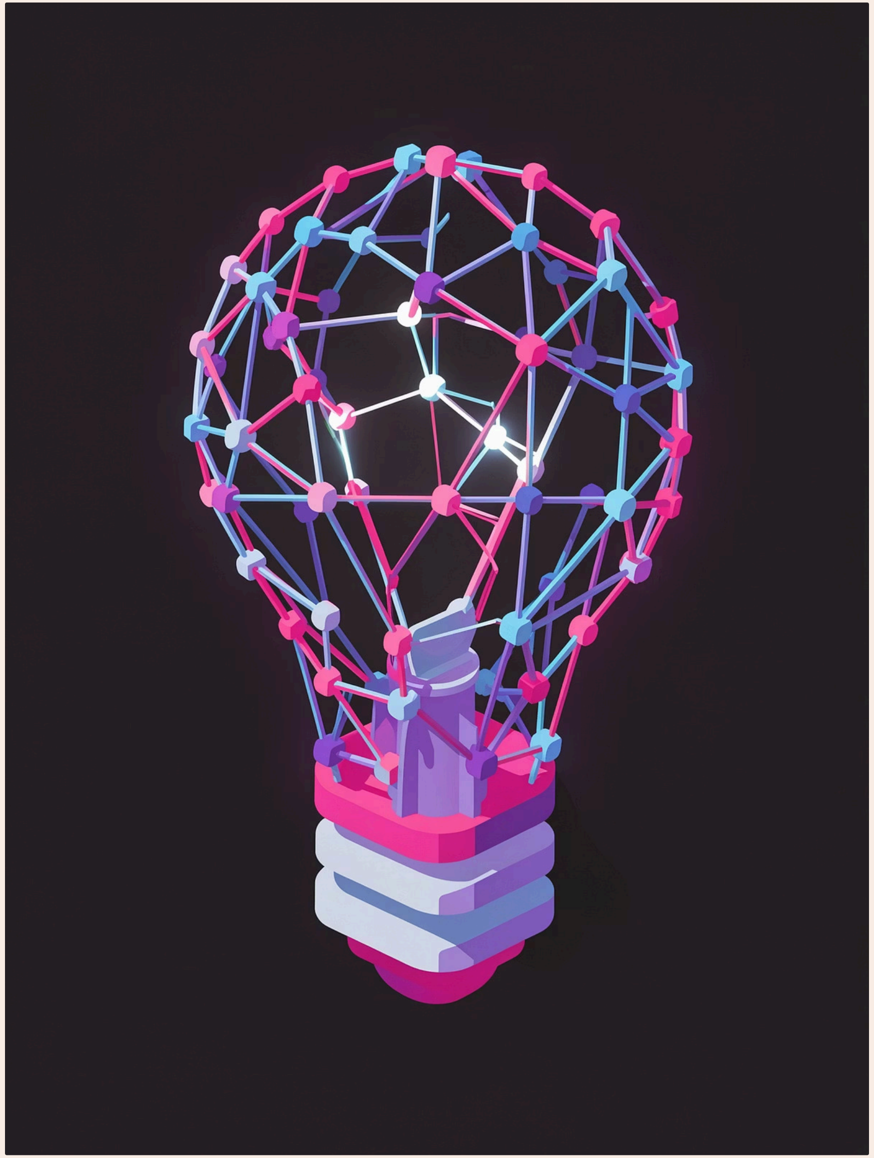
DDPM — This Project

- 300 denoising steps
- 300 U-Net forward passes per image

SDXL-Turbo — Part I

- Distilled from a DDPM-like teacher
- 1 forward pass, 0.79s — **13× faster**

Key insight: One-step distillation cannot exist without a well-trained DDPM teacher. The quality ceiling of LCM, DMD2, and SDXL-Turbo is set by how well the teacher was trained.



What Building from Scratch Revealed

1. **Reparameterization Trick is Essential**

   Without it, training would require 300 sequential forward passes per sample — making DDPM computationally intractable.

2. **Simplified MSE Loss Works Surprisingly Well**

   Despite ignoring the full variational lower bound, the simplified objective produces high-quality samples.

3. **EMA on Weights is Non-Negotiable**

   Without EMA, the checkerboard artifact dominates at 32×32. EMA is essential for visual quality.

References

1

Ho et al. — DDPM

"Denoising Diffusion Probabilistic Models." NeurIPS 2020. arXiv:2006.11239

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"Consistency Models." ICML 2023. arXiv:2303.01469

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Sauer et al. — Adversarial Diffusion Distillation

"Adversarial Diffusion Distillation." arXiv:2311.17042

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Yin et al. — Improved Distribution Matching Distillation

"Improved Distribution Matching Distillation." NeurIPS 2024. arXiv:2405.14867

Dataset: Krizhevsky, A. (2009). CIFAR-10. University of Toronto.